MODULE 1 · LECTURE 30

Digital India and Initiatives for Agricultural Digitalization

The Digital India programme and its vision areas, the digital public infrastructure and connectivity that farmers rely on, the timeline of India's digital-agriculture initiatives, the Digital Agriculture Mission of 2024, e-NAM as a digital market, data protection, and the lessons on what makes digital services work for farmers.

Learning objectives

After this lecture you should be able to:

1. Describe the Digital India programme and its three vision areas.
2. Explain digital public infrastructure and connectivity initiatives that reach farmers.
3. Trace the main digital-agriculture initiatives in India and their purposes.
4. Describe the components of the Digital Agriculture Mission.
5. Explain how a digital market such as e-NAM can change price discovery, using a simulation.
6. Discuss data protection and the conditions under which digital services benefit farmers.

Scope: earlier lectures described individual services — e-NAM and Kisan Call Centre (Lecture 7), Common Service Centres and PM-KISAN (Lecture 10), farm apps (Lecture 14), YES-TECH (Lecture 26) and Pashu Aadhaar (Lecture 24). This lecture places them within the national Digital India programme and the Digital Agriculture Mission. AgriStack and the India Digital Ecosystem of Agriculture are examined in depth in Lecture 31.

1. Digital India

Digital India is a flagship programme of the Government of India, launched on 1 July 2015, with the vision of transforming India into a digitally empowered society and knowledge economy, through digital delivery of services and the expansion of the digital economy and employment (Digital India, n.d.).

Digital India (launched 1 July 2015): three vision areas

Digital infrastructure as a utility to every citizen	Governance and services on demand	Digital empowerment of citizens
high-speed internet lifelong digital identity mobile phone + bank account Common Service Centre access cloud space; safe cyberspace	services integrated across departments; online and on mobile in real time; electronic, cashless payments; GIS for decisions	universal digital literacy; resources and services in Indian languages; participative governance; portable entitlements
For farmers *BharatNet fibre to the panchayat; Aadhaar-linked bank account for DBT*	**For farmers** *apply for schemes online; Soil Health Card, PMFBY and PM-KISAN portals*	**For farmers** *advisories and e-NAM in local languages; digital literacy training*

Vision areas and their components summarised from the Digital India website; farm examples are original.

Figure 1. The three vision areas of Digital India and what they mean for farmers.

The first vision area treats high-speed internet as a **core utility**, like water or electricity, and adds a lifelong digital identity, a mobile phone and bank account for every citizen, easy access to a Common

Service Centre, and private space on a public cloud (Digital India, n.d.). Together, identity, payments and data-sharing systems built for the whole population are called **digital public infrastructure (DPI)**; agriculture now builds its own DPI on top of them (Section 4).

Building block	What it provides	Use in agriculture
Digital identity (Aadhaar)	A unique, verifiable identity — more than 144 crore Aadhaar numbers generated (Digital India, n.d.)	Identifying beneficiaries; linking Farmer ID (Lecture 31)
Bank account and digital payments	Direct transfers and instant payments; the BHIM payments app was launched in 2016	Direct benefit transfer such as PM-KISAN; e-NAM payments
Service points	Common Service Centres run by village entrepreneurs (Lecture 10)	Online applications for farmers without devices
Connectivity	BharatNet fibre and the National Broadband Mission (Section 2)	Access to apps, advisories and markets
Digital literacy	Pradhan Mantri Gramin Digital Saksharta Abhiyan (PMGDISHA), launched in 2018	Farmers able to use digital services safely

Worked example 1 — direct benefit transfer at scale
PM-KISAN pays eligible farmer families ₹6,000 a year in three instalments (Lecture 10).
The 21st instalment transferred ₹18,000 crore to 9 crore farmers: 18,000 crore ÷ 9 crore = **₹2,000 per farmer** (Press Information Bureau [PIB], 2025b).
For a village with 400 eligible families: 400 × ₹6,000 = **₹24 lakh a year**, paid straight into bank accounts without intermediaries — possible only because identity, bank accounts and payments are digital.

2. Connectivity and the digital divide

BharatNet lays optical fibre cable to gram panchayats. Of about 2,56,000 gram panchayats, nearly 2,14,000 had been made online under BharatNet Phases I and II at a cost of about ₹42,000 crore; the remaining panchayats and non-panchayat villages are covered under the Amended BharatNet Programme. India's broadband subscribers grew from 61 million to over 1 billion, with an average fixed broadband speed of 61.55 Mbps. The **National Broadband Mission 2.0**, launched on 1 April 2025, targets fibre connectivity with 95% uptime in 2.7 lakh villages (42,000 had it in December 2025) and an average speed of 100 Mbps by 2030 (PIB, 2026b).

Worked example 2 — how far is universal connectivity?
Share of gram panchayats online = 2,14,000 ÷ 2,56,000 = **83.6%**; about 42,000 remain.
National Broadband Mission 2.0: from 42,000 villages with reliable fibre (December 2025) to 2.7 lakh by 2030 needs (2,70,000 − 42,000) ÷ 5 ≈ **45,600 villages a year**.
Connection to the panchayat is only the first step: the last mile to farms and homes, affordable devices and the skills to use them decide whether farmers benefit.

3. India's digital-agriculture initiatives

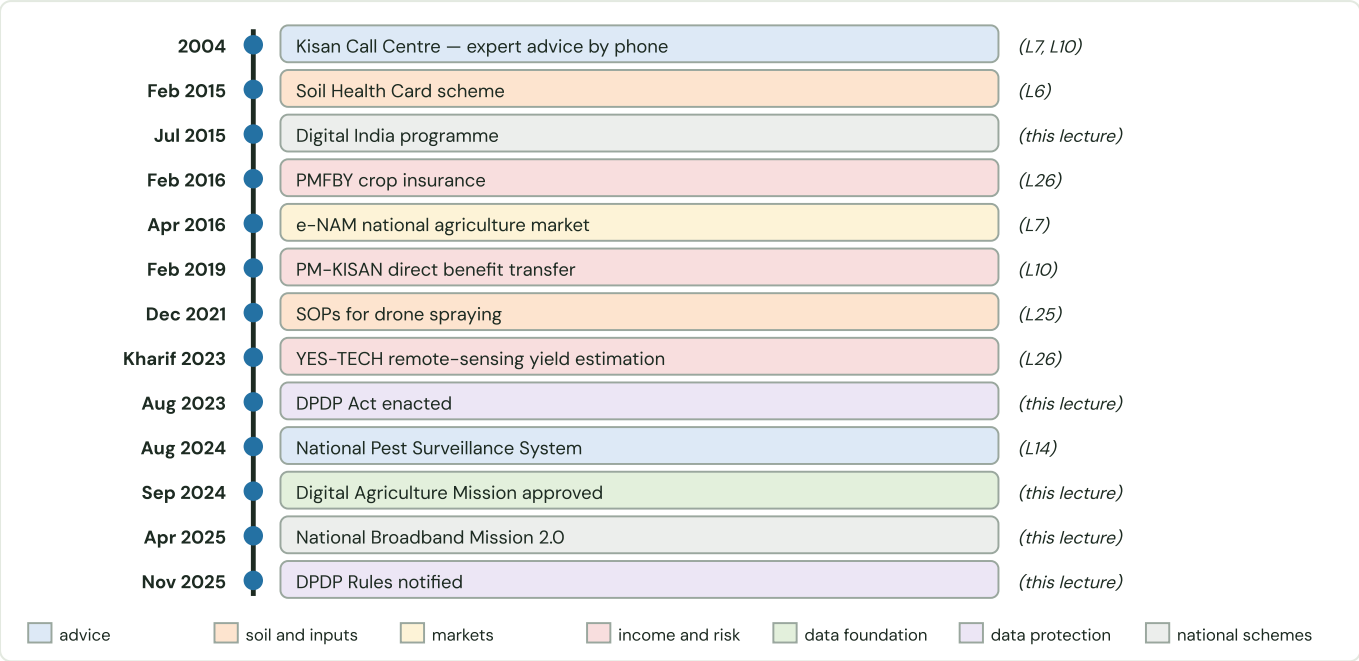


Figure 2. Timeline of selected digital initiatives for agriculture, with the lecture in which each is discussed. Dates are from sources cited in this and earlier lectures. Original diagram.

Purpose	Initiatives	Lecture
Advice and information	Kisan Call Centre (2004), mKisan SMS advisories, Kisan Suvidha and Meghdoot apps, National Pest Surveillance System (2024)	7, 10, 14, 25
Soil and inputs	Soil Health Card scheme, launched on 19 February 2015 at Suratgarh, Rajasthan (PIB, 2020); soil profile mapping	6, this lecture
Markets	e-NAM (April 2016), AGMARKNET price data, electronic negotiable warehouse receipts	7, 14, this lecture
Income support and risk	PM-KISAN (2019) direct benefit transfer; PMFBY crop insurance, launched on 18 February 2016 (PIB, 2026c), with YES-TECH yield estimation (2023)	10, 26
Livestock	Bharat Pashudhan and Pashu Aadhaar ear tags	24
Mechanisation and inputs	SOPs for drone spraying (2021) and crop-specific SOPs (2023)	25
Data foundation	Digital Agriculture Mission (2024): AgriStack, Krishi DSS, soil profile maps, digital crop surveys	this lecture, 31

4. The Digital Agriculture Mission (2024)

On 2 September 2024 the Union Cabinet approved the **Digital Agriculture Mission** with an outlay of ₹2,817 crore, including a central share of ₹1,940 crore. It is an umbrella scheme for creating digital public infrastructure for agriculture, implementing the Digital General Crop Estimation Survey and supporting other IT initiatives of the Centre, States and research institutions (PIB, 2024).

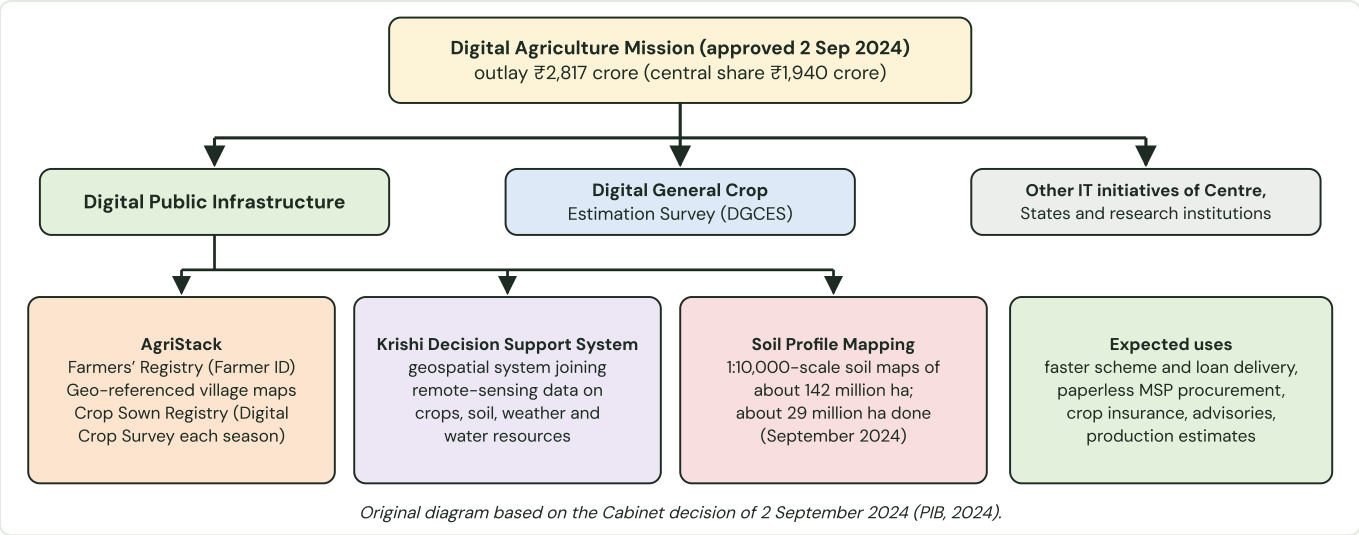


Figure 3. Structure of the Digital Agriculture Mission. Original diagram.

- **AgriStack** is a farmer-centric DPI built in a federated way by the Centre and States. Its three foundational registries — the Farmers' Registry, geo-referenced village maps and the Crop Sown Registry — are created and maintained by State Governments. Each farmer gets a **Farmer ID**, a trusted "Kisan ki Pehchaan", linked to land records, crops sown, livestock and benefits availed; the target was 11 crore Farmer IDs by 2026–27. Crops sown are recorded each season through mobile-based **Digital Crop Surveys** (PIB, 2024).
- The **Krishi Decision Support System** is a geospatial system that unifies remote-sensing information on crops, soil, weather and water resources (PIB, 2024).
- **Soil profile maps** at 1:10,000 scale are planned for about 142 million ha of agricultural land; about 29 million ha had been completed (PIB, 2024).
- The **Digital General Crop Estimation Survey** provides yield estimates from scientifically designed crop-cutting experiments; with digitally recorded crop area and remote sensing it supports accurate production estimates (PIB, 2024; Lecture 26).
- Digital crop surveys and ground-truth data collection were expected to employ about 2.5 lakh trained local youth and Krishi Sakhis (PIB, 2024).

The Mission expects that farmers will be able to identify themselves digitally to obtain schemes, crop loans, advisories and market connections with little paperwork, and that trusted data will make procurement at MSP, crop insurance and credit-card-linked crop loans more efficient and transparent (PIB, 2024). How AgriStack works and the principles of the India Digital Ecosystem of Agriculture are the subject of Lecture 31.

5. Worked example: e-NAM as a digital market

e-NAM, launched in April 2016, digitally integrates existing mandis into a national electronic trading platform. Its portal covers the whole trade cycle — digital gate entry, a unique lot ID, quality assaying, competitive online bidding with real-time tracking, price discovery and direct electronic payment to the farmer's bank account; a State Unified License lets traders bid across state boundaries, and the interface works in 12 languages. By March 2026 it had registered over 1.80 crore farmers, 2.73 lakh traders and 4,724 FPOs (PIB, 2026a).

Why might more bidders help farmers? The program simulates 300 wheat lots that are assayed and offered to 4 local traders, then also to 4, 8 or 16 traders from other mandis who must pay transport. In

open bidding, the price ends just above the second-highest trader's valuation. All values are illustrative; the program shows the mechanism, not measured e-NAM results.

Simulating online bidding with more traders in Python (Lecture 8)

```
import random

random.seed(30)
BASE = 2450                                  # typical value of FAQ wheat, Rs per quintal
(illustrative)
LOTS = 300                                   # farmers' lots, 25 quintals each

# Each lot is assayed (quality 0.95-1.05). Every trader values it slightly differently
# (their own buyers and needs); outside traders must also pay to move the grain.
lots = []
for _ in range(LOTS):
    q = random.uniform(0.95, 1.05)
    local = [random.gauss(1.0, 0.03) for _ in range(4)]
    outside = [(random.gauss(1.0, 0.03), random.random()) for _ in range(16)]
    lots.append((q, local, outside))

def auction(n_outside, transport=(60, 150)):
    """Open online bidding: the price ends just above the second-highest valuation."""
    lo, hi = transport
    total, won_outside = 0.0, 0
    for q, local, outside in lots:
        bids = [(BASE * q * f, "local") for f in local]
        bids += [(BASE * q * f - (lo + u * (hi - lo)), "outside") for f, u in outside[:n_outside]]
        bids.sort(reverse=True)
        total += bids[1][0]                  # price paid by the winner
        won_outside += bids[0][1] == "outside"
    return total / LOTS, 100 * won_outside / LOTS

base, _ = auction(0)
print("traders bidding                        price (Rs/q)  won by outside  gain per 25 q lot")
for n_out, cost in [(0, (60, 150)), (4, (60, 150)), (8, (60, 150)), (16, (60, 150)), (16, (20, 60))]:
    price, share = auction(n_out, cost)
    label = f"4 local + {n_out:2d} outside" + (", cheap transport" if cost == (20, 60) else "")
    print(f"{label:38s} {price:6.0f}        {share:5.1f}%        Rs {25 * (price - base):5.0f}")

Output:
traders bidding                       price (Rs/q)  won by outside  gain per 25 q lot
4 local +  0 outside                     2477          0.0%        Rs     0
4 local +  4 outside                     2483          9.0%        Rs   158
4 local +  8 outside                     2488         12.7%        Rs   275
4 local + 16 outside                     2496         22.0%        Rs   474
4 local + 16 outside, cheap transport    2525         57.3%        Rs  1212
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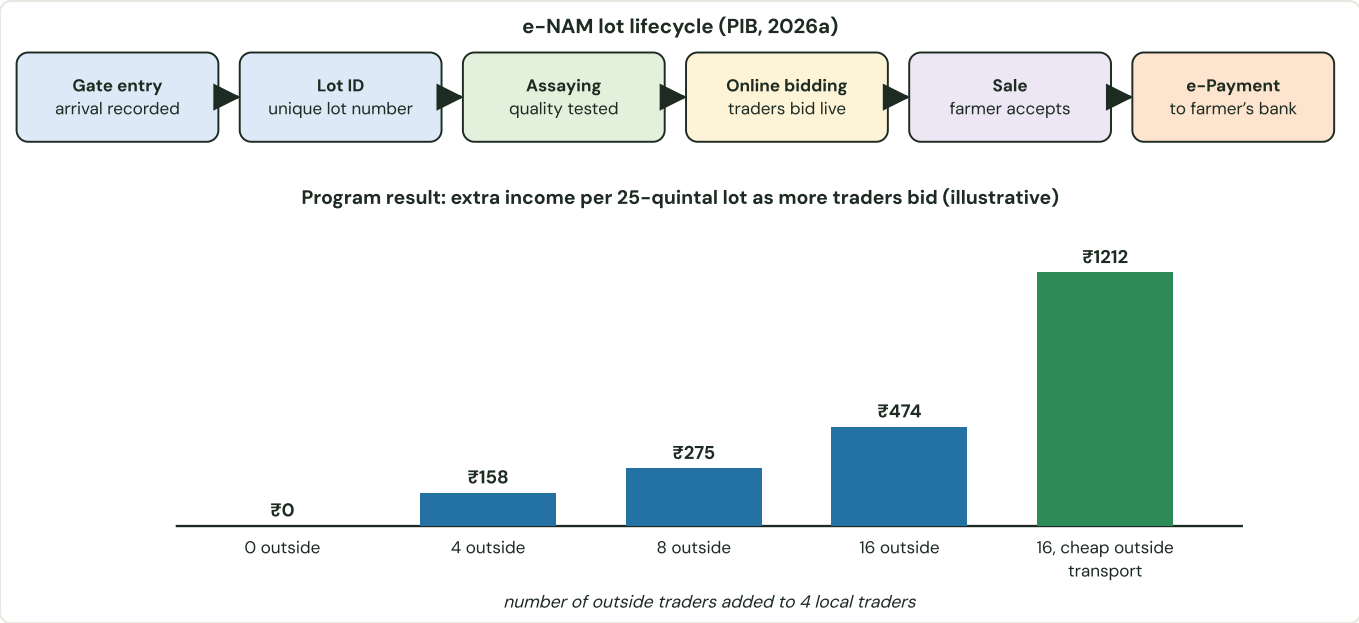


Figure 4. Top: the e-NAM trading steps. Bottom: extra income per lot in the simulation. Original diagram and chart.

- More bidders raise the price: adding 4, 8 and 16 outside traders increases income per 25-quintal lot by about ₹158, ₹275 and ₹474 — about ₹6, ₹11 and ₹19 per quintal.
- Outside traders win only 9–22% of lots, because transport costs of ₹60–150 per quintal handicap them; yet their bids still push local traders to pay more.
- When transport becomes cheaper (₹20–60 per quintal), outside traders win 57% of lots and the gain rises to ₹1,212 per lot: digital markets pay off fully only with good roads, logistics, trusted assaying and prompt payment.

The program was run and the output shown is its actual output. Prices, costs and trader behaviour are illustrative assumptions, not e-NAM data.

6. Data protection

Digital agriculture collects personal data on farmers, their land, crops, loans and benefits. The **Digital Personal Data Protection (DPDP) Act**, enacted on 11 August 2023, sets the obligations of organisations that handle personal data (Data Fiduciaries) and the rights and duties of individuals (Data Principals). Its seven core principles are consent and transparency, purpose limitation, data minimisation, accuracy, storage limitation, security safeguards and accountability. The **DPDP Rules, 2025**, notified in November 2025, require clear, standalone consent notices, prompt notice of data breaches, and responses within 90 days to requests to access, correct, update or erase personal data, with an 18-month phased compliance timeline (PIB, 2025a).

- A farmer should know **why** data are collected and agree to each use — for example, sharing crop data with a lender only with consent.
- Collect only what is needed, keep it accurate, and protect it from leaks.
- Farmers must be able to see and correct their records — errors in land or crop data can deny them benefits.

7. What makes digital services work for farmers?

Information and communication technologies have created many new services for farmers — over 140 deployments worldwide in 2015 — but evidence on their impact on adoption, behaviour and welfare is mixed. Users may not engage with services whose design does not suit them, and trust in the source of

information matters; designers should first understand whether lack of information is really the farmer's binding constraint (Aker et al., 2016).

Challenge	Why it matters	Response
Connectivity and devices	Villages without reliable internet or smartphones are left out	BharatNet, NBM 2.0; Common Service Centres; SMS and voice services
Digital literacy and language	Complex apps in English exclude many farmers	Training (PMGDISHA); local languages, voice, simple design
Data quality	Wrong land or crop records deny benefits	Digital crop surveys, ground truth, easy correction
Inclusion	Tenant farmers and women may lack land titles	Registries that include cultivators and tenants as per State policy (PIB, 2024)
Trust and privacy	Farmers will not share data they fear will be misused	Consent, purpose limitation, DPDP safeguards
Complementary services	Information alone does not raise income	Roads, storage, credit, assaying and payment systems (Section 5)

8. Key takeaways

- Digital India (1 July 2015) aims at a digitally empowered society through infrastructure as a utility, services on demand and digital empowerment.
- Identity, bank accounts, payments, service centres, connectivity and literacy form the digital public infrastructure farmers use.
- About 83.6% of gram panchayats were online under BharatNet; NBM 2.0 targets reliable fibre in 2.7 lakh villages by 2030.
- Digital agriculture grew from Kisan Call Centre (2004) through Soil Health Card, e-NAM, PMFBY and PM-KISAN to the Digital Agriculture Mission (2024).
- The Digital Agriculture Mission builds AgriStack, the Krishi Decision Support System and soil profile maps, plus the Digital General Crop Estimation Survey.
- e-NAM digitises the whole trade cycle; more bidders raise prices, but only with good logistics, assaying and payment.
- The DPDP Act 2023 and Rules 2025 require consent, purpose limitation and the right to correct data — vital for farmer data.

9. Examination questions

Short-answer questions (2 marks)

Q1. When was Digital India launched and what is its vision?

1 July 2015; to transform India into a digitally empowered society and knowledge economy.

Q2. Name the three vision areas of Digital India.

Digital infrastructure as a utility to every citizen; governance and services on demand; digital empowerment of citizens.

Q3. What is digital public infrastructure?

Shared digital systems — identity, payments, data registries — built for the whole population and used by many services.

Q4. What is BharatNet?

A programme laying optical fibre cable to connect gram panchayats to broadband.

Q5. State two targets of the National Broadband Mission 2.0.

Reliable fibre (95% uptime) in 2.7 lakh villages and an average fixed broadband speed of 100 Mbps by 2030.

Q6. When and where was the Soil Health Card scheme launched?

On 19 February 2015 at Suratgarh, Rajasthan.

Q7. Name the three DPIs of the Digital Agriculture Mission.

AgriStack, Krishi Decision Support System and Soil Profile Mapping.

Q8. Name the three registries of AgriStack.

Farmers' Registry, geo-referenced village maps and Crop Sown Registry.

Q9. List the steps of trading a lot on e-NAM.

Gate entry, lot ID, assaying, online bidding, sale and e-payment to the farmer.

Q10. Give four principles of the DPDP Act, 2023.

Any four: consent and transparency, purpose limitation, data minimisation, accuracy, storage limitation, security safeguards, accountability.

Long-answer questions (5–10 marks) — answer outlines

Q1. Describe the Digital India programme and its relevance to agriculture.

Launch and vision; Figure 1; DPI table; worked example 1.

Q2. Discuss connectivity initiatives and the digital divide in rural India.

BharatNet progress; NBM 2.0 targets; worked example 2; last mile, devices, skills.

Q3. Trace the development of digital-agriculture initiatives in India.

Figure 2 timeline; purpose table; links to earlier lectures.

Q4. Explain the components and expected benefits of the Digital Agriculture Mission.

Cabinet decision; Figure 3; AgriStack registries and Farmer ID; Krishi DSS; soil maps; DGCES; uses.

Q5. Explain how e-NAM works and how online bidding can affect farmers' prices.

Lot lifecycle; statistics; program and Figure 4; competition and logistics.

Q6. Discuss data protection and the conditions for digital services to benefit farmers.

DPDP Act and Rules; principles for farmer data; Aker et al. (2016); challenges table.

10. Glossary

Term	Meaning
AgriStack	Farmer-centric digital public infrastructure with three registries
Assaying	Testing the quality of a lot of produce
BharatNet	Optical-fibre network connecting gram panchayats
Consent notice	Clear statement of why personal data are collected
Data Fiduciary	Organisation that decides how personal data are processed
Data Principal	Person to whom personal data relate
DBT	Direct benefit transfer into beneficiaries' bank accounts
DGCES	Digital General Crop Estimation Survey
Digital Crop Survey	Mobile-based seasonal recording of crops sown
Digital divide	Gap in access to digital technology between groups
DPI	Digital public infrastructure: shared identity, payment and data systems
e-NAM	National Agriculture Market electronic trading platform
Farmer ID	Digital identity of a farmer under AgriStack
Krishi DSS	Geospatial Krishi Decision Support System
NBM 2.0	National Broadband Mission 2.0 (2025)
Purpose limitation	Using personal data only for the stated purpose

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All diagrams, examples and programs are original. Figures and dates are from the official sources cited, as accessed on 9 November 2026; check the latest releases before quoting them. Prices and costs in the program are illustrative.

Practice MCQs

Q1. The Digital India programme was launched on

(a) 15 August 2014 (b) 1 July 2015
(c) 26 January 2017 (d) 2 September 2024

Q2. Which is NOT one of the three vision areas of Digital India?

(a) Digital infrastructure as a utility to every citizen (b) Governance and services on demand
(c) Digital empowerment of citizens (d) Free smartphones for every farmer

Q3. BharatNet provides

(a) satellite TV to villages (b) optical-fibre connectivity to gram panchayats
(c) free electricity (d) crop insurance

Q4. About what share of India's gram panchayats had been made online under BharatNet Phases I and II?

(a) 25% (b) 50%
(c) 84% (d) 100%

Q5. The National Broadband Mission 2.0 targets reliable fibre connectivity by 2030 in about

(a) 42,000 villages (b) 2.7 lakh villages
(c) 10 lakh villages (d) 2 crore villages

Q6. The Soil Health Card scheme was launched in 2015 at

(a) Pune (b) Suratgarh, Rajasthan
(c) New Delhi (d) Nagpur

Q7. The Digital Agriculture Mission approved in September 2024 has an outlay of

(a) ₹281.7 crore (b) ₹2,817 crore
(c) ₹28,170 crore (d) ₹2.8 lakh crore

Q8. Which is one of the three foundational registries of AgriStack?

(a) Crop Sown Registry (b) Vehicle Registry
(c) Voter Registry (d) Patent Registry

Q9. The Krishi Decision Support System is mainly a

(a) call centre (b) geospatial system unifying remote-sensing data on crops, soil, weather and water
(c) fertiliser subsidy (d) tractor database

Q10. In e-NAM, quality testing of a lot before bidding is called

(a) assaying (b) winnowing
(c) gate entry (d) e-payment

Q11. In the e-NAM simulation, the gain to farmers was largest when

(a) no outside traders bid (b) many outside traders bid and transport was cheap
(c) only local traders bid (d) lots were not assayed

Q12. The DPDP Act was enacted by Parliament on

(a) 11 August 2023 (b) 1 July 2015

(c) 14 April 2016 (d) 2 September 2024

Answers

Q1 — (b) 1 July 2015

Q2 — (d) Free smartphones for every farmer

Q3 — (b) optical-fibre connectivity to gram panchayats

Q4 — (c) 84%

Q5 — (b) 2.7 lakh villages

Q6 — (b) Suratgarh, Rajasthan

Q7 — (b) ₹2,817 crore

Q8 — (a) Crop Sown Registry

Q9 — (b) geospatial system unifying remote-sensing data on crops, soil, weather and water

Q10 — (a) assaying

Q11 — (b) many outside traders bid and transport was cheap

Q12 — (a) 11 August 2023